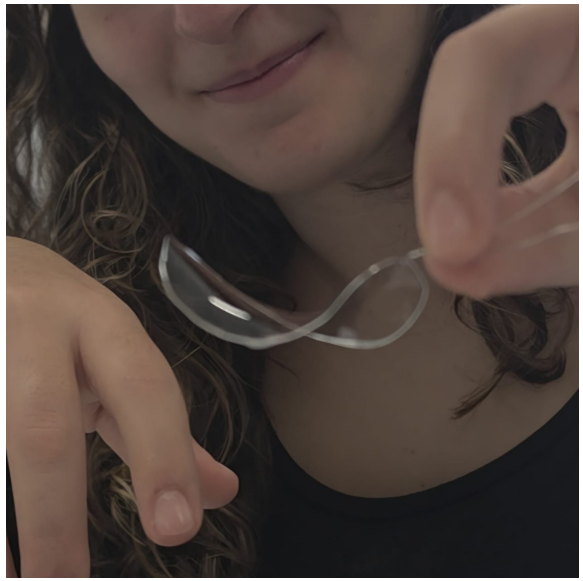
Bernstein's problem

Eugenelo Tiziana
Supervisor: Guido De Philippis

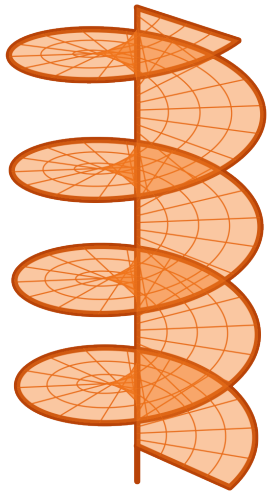
October 23rd

Plateau Problem

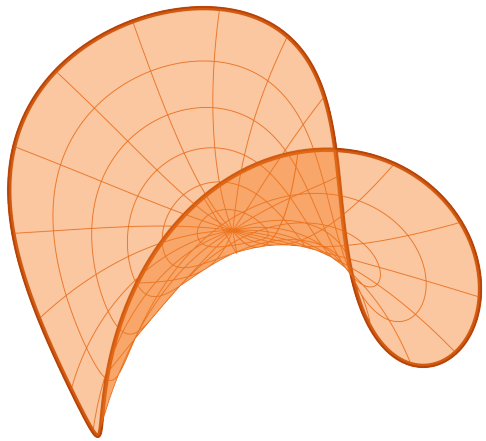




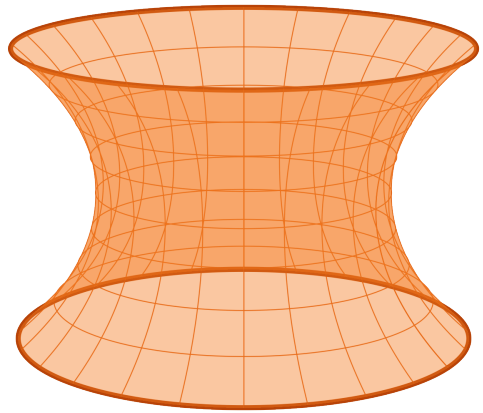




Elicoid

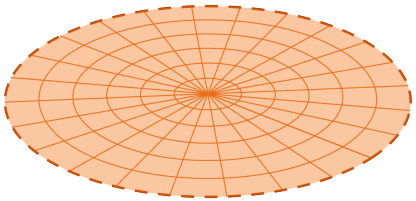


Saddle

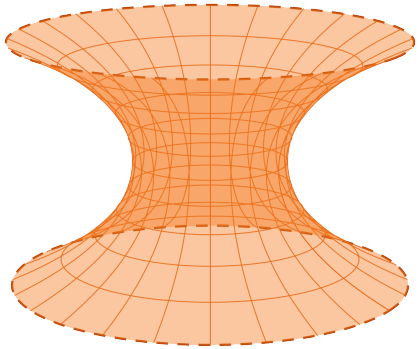


Catenoid

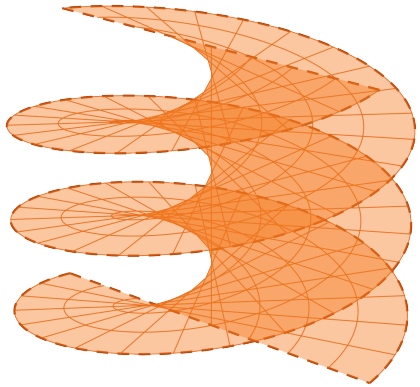
Complete Surfaces



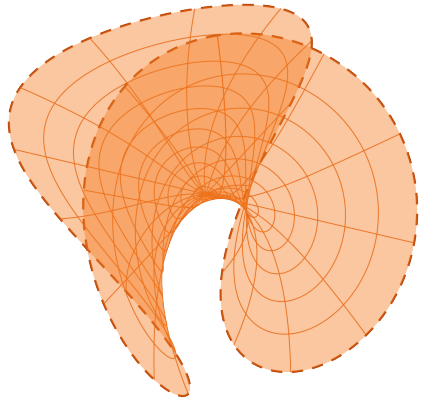
Plane



Catenoid



Elicoid



Enneper

We work in codimension 1

Critical points

We consider functions $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$. Then the area of Graph$(f)$ is defined as

$$A(f) = \int_\Omega |J_{(x,f(x))}| dx = \int_\Omega \sqrt{1 + |\nabla f|^2}.$$

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We can avoid flatness if the curvature balances out.

Harmonicity

Note that if you assume $|Df| << 1$, then

$$\operatorname{div}\Big(\frac{\nabla f}{\sqrt{1+|\nabla f|^2}}\Big) \approx \operatorname{div}(\nabla f) = \Delta f = 0$$

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The area looks like an harmonic function. Its second fundamental form A looks like its Hessian, and if $A = (a_{ij})_{i,j\in\{1,\dots,n\}}$.

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We have a lot of rigidity (!), and regularity (!!).

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Theorem (Liouville). Let $u : \mathbb{R}^n \to \mathbb{R}$ harmonic. If u has bounded gradient, it is affine.

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Yes, for entire graphs with bounded gradient (Moser, 1961).
No, for manifolds (the catenoid!)

Can we remove the hypothesis on the gradient if we restrict to a smaller class?

Stability inequality

A critical point for the area f is stable if $\forall \phi \in C_c^\infty(\Omega)$

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You can see this as an integral bound on the curvature.

Stable Bernstein problem

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Counterexample (Bombieri, De Giorgi, Giusti). In $\mathbb{R}^8$,

$$C = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 | |x| = |y|\}$$

is *area-minimizing*. We can use this to build a counterexample for $u : \mathbb{R}^8 \to \mathbb{R}$ with has area-minimizing graph.

Monotonicity formula

If $M \subset \mathbb{R}^n$ is a minimal hypersurface, then $\forall x \in M$,

$$\frac{\mathcal{H}^{n-1}(B_r(x) \cap M)}{r^{n-1}}$$

is monotone in $r \in \mathbb{R}_{>0}$.

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The monotonicity formula gives us the conic structure infinitesimally.

Tools

Following Schoen-Simon-Yau (1975), we want to leverage the properties of minimal surfaces to learn as much as possible on A and ∇A, as we want to test by A localized.

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We are going to leverage the harmonic structure, by the two following tools

- Derivative of harmonic functions are harmonic
- Bochner formula

Structure of ∇A

For harmonic functions, we get $\Delta \partial_i u = \partial_i \Delta u = 0$.

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Taking one derivative isn't too annoying.

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By which

$$\frac{1}{2}\Delta |Du|^2 = \operatorname{div}(D^2 u\, Du) = |D^2 u|^2 + \cancel{\langle Du, D\Delta u \rangle}$$

Simons' identity

For the second fundamental form we have

$$\Delta A = -|A|^2 A$$

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which gives Simons' identity

$$|A|\Delta|A| + |\nabla|A||^2 = |\nabla A|^2 - |A|^4.$$

$|\nabla A|$ and $|\nabla |A||$

We now need something that can bind $|\nabla A|^2$ and $|\nabla |A||^2$ if we want to use Simons and stability.

This means being able to understand how the change of A and $|A|$ relate, so it brings us back to its tensorial structure.

Kato inequality-1

By Cauchy-Shwarz

$$\nabla_k |A| = \frac{\langle \nabla_k A, A \rangle}{|A|} \leq \frac{|\nabla_k A||A|}{|A|},$$

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This is true for any tensor. To get something stronger we want to leverage the traces and the symmetry. How much more can we gain?

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The whole proof reduces to the use of the symmetry and nullity of the traces of A and ∇A.

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&= 2\sum_{i,j} a_{ij}a_{ijk} \\
&= 2\sum_{i} \lambda_i a_{iik}.
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&= S + T
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This gives

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$$\leq (n-1) \sum_k \sum_{i \neq k} a_{iik}^2$$
$$= (n-1) T .$$

Kato inequality - 5

And lastly

$$|\nabla A|^2 = \sum_i a_{iii}^2 + \sum_{i,j} a_{iij}^2 + a_{iji}^2 + a_{jii}^2 + \sum_{i \neq j \neq k \neq i} a_{ijk}^2$$

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By putting everything together:

$$|\nabla |A||^2 \leq S + T, \quad |\nabla A|^2 \geq S + 3T, \quad S \leq (n-1)T,$$

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we get

$$|\nabla A|^2 \geq \left(1 + \frac{2}{n}\right)|\nabla |A||^2$$

Refined Kato inequality! Due to Schoen-Simon-Yau (1975)

Simons and Kato

By combining Simons' formula and the refined Kato inequality, i.e.

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Let's see what we can find out thanks to this.

Schoen-Simon-Yau

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Theorem (Schoen-Simon-Yau). For $M^n \subset \mathbb{R}^{n+1}$, for $q < 2/n$, we get

$$C(n,p)\int_M |A|^{2q+2}\eta^{2q+2} \leq \int_M |\nabla\eta|^{2q+2},$$

where $C(n,p) > 0$ for $q < 1 + \sqrt{2/n}$.

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If we use this to localize?

Growth bound

By choosing η bump on $B_{\theta r}(x), B_r(x)$, we get

$$\int_{B_{\theta r}(x)\cap M} |A|^{2q+2} \leq \frac{\tilde{C}(n,q,\theta)}{r^{2q+2}} |B_r|$$

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$$\int_{B_{\theta r}(x)\cap M} |A|^{2q+2} \leq \frac{\tilde{C}(n,q,\theta)\Lambda}{r^{2q+2-n}}$$

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If we combine with $q < 1+\sqrt{2/n}$, we get $n \leq 5$.

Bellettini

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Theorem (Bellettini). There is $\epsilon_0 = \epsilon_0(n)$. Let M be a properly immersed two-sided stable minimal hypersurface in $B^{n+1}_{2R}(0)$, with $0 \in M$ and such that

$$\frac{1}{(2R)^{n-2}} \int_{M \cap B^{n+1}_{2R}(0)} |A_M|^2 \leq \epsilon_0,$$

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then, we have for some explicit $c = c(n)$

$$\sup_{M \cap B^{n+1}_{R/2}(0)} |A_M| \leq \frac{c(n)}{R} \Big(\frac{1}{(2R)^{n-2}} \int_{M \cap B^{n+1}_{2R}(0)} |A_M|^2 \Big)^{1/2}.$$

ϵ-regularity

The proof is obtained by an an iteration scheme, similar to the one of De Giorgi-Nash-Moser theorem. We first obtain an intrinsic Caccioppoli inequality: for any $k > 0$, $\eta \in C^1_C$

$$\int_{\{|A|>k\}} \left(1 - \frac{k}{|A|}\right) |\nabla |A||^2 \eta^2 \leq \frac{1}{c} \int \left((|A| - k)^+\right)^2 |\nabla \eta|^2$$
$$+ \frac{k}{c} \int \left((|A| - k)^+\right)^3 \eta^2 + \frac{2k^2}{c} \int \left((|A| - k)^+\right)^2 \eta^2$$
$$+ \frac{k^3}{c} \int (|A| - k)^+ \eta^2.$$

Obtained by testing with $(|A| - k)^+ \eta$, (we use Simons' formula and Kato to cancel the term $|A|^4$).

Michael-Simon inequality

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Theorem (Michael-Simon inequality). Let $M^n \subset \mathbb{R}^n$ an immersed minimal surface. Then $\forall f \in C_c^\infty(M)$,

$$\Big(\int_M |f|^{\frac{n}{n-1}} \Big)^{\frac{n-1}{n}} \leq C(n) \int_M |\nabla f|.$$

De Giorgi iteration

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$$k_i = d - d/2^{i-1}, \qquad R_i = R/2 + R/2^i,$$

and

$$S_i = \int_{B_{R_i}} ((|A| - k_i)^+)^{\frac{2n}{n-2}}$$

The risk is that the function is highly concentrated, but the Caccioppoli inequality and Michael-Simon inequality prevent that.

De Giorgi iteration - 2

We get that

$$S_{i+1} \leq C(n) R^{\alpha(n)} S_{i-1}^{1+\frac{2}{n-2}}$$

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It's superlinear so we're okay. This forces $S_i \to 0$ as long as S_1 is small enough, which is true for

$$\varepsilon_0 = \frac{1}{C_{MS}^{1+\frac{n-2}{2}} \, 4(n+2) \, 2^{n-2} \, (32n)^{\frac{n-2}{2}} \, 2^{\frac{n(3n-2)}{2}}}$$

(this is why I won't show the calculations)

What we are missing is convincing ourselves that for all our surfaces we get

$$\frac{1}{(2R)^{n-2}} \int_{M \cap B^{n+1}_{2R}(0)} |A_M|^2 \leq \epsilon_0,$$

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This result is achieved by using both weak and strong objects, one for the smoothness and curvature, the other for the singularities (!).

Removing the growth Bound

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- $n = 6$ Hong-Li-Wang, 2026.

Yes!

$n = 2$

We have that A has the following structure,

$$A = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}.$$

which means that

$$K = \det(A) = -\frac{1}{2}|A|^2 \leq 0$$

with K Gaussian curvature of M.

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The stability inequality becomes

$$\int_M 2K\phi^2 + |\nabla\phi|^2 \geq 0$$

$n = 2$

Theorem (Fischer-Colbrie-Schoen). If $M \subset \mathbb{R}^3$ is an oriented complete stable minimal 2-manifold, then M is conformally equivalent to the complex plane $\mathbb{C}$ or the cylinder A.

In either case we can lift to a complex universal covering $\mathbb{C}$ of M. This will force the flatness.

$n = 3$

Theorem (Chodosh-Li). A complete, two-sided, minimal, stable immersion $M^3 \to \mathbb{R}^4$ has finite total curvature

$$\int_M |A_M|^3 < \infty$$

We can actually ask a bit less than stability. The following corollary follows (with a lot of work)

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We can actually ask a bit less than stability. The following corollary follows (with a lot of work)

Corollary. Suppose M is as above. Then, M has cubic volume growth, i.e.:

$$\sup_{r \to \infty} \frac{\mathrm{Vol}(B_r(0) \cap M)}{r^3} < \infty.$$

$n = 3$

The case $n = 3$ is settled by working with Green functions, i.e. a function u such that

$$-\Delta_y u(x, y) = \delta_y$$

and showing that if we take

$$F(t) := \int_{u^{-1}(t)} |\nabla u|^2$$

and prove $F(t) = O(t^2)$; then we get our growth bound. The point is to work on level sets and not balls, and we want some nice level sets that look like balls.



Thank you for the attention!